

Practical Assignment Sheet

1. Perform following tasks:
 - 1.1. Create database named 'assignment.db'
 - 1.2. Create table named 'tblsub_master' with fields 'sid, sub_name, sem, credit'
 - 1.3. Create table named 'tblteacher_master' with fields 'tid, name, gender, city'
 - 1.4. Create table named 'tblstud_master' with fields 'rno, name, sem'
 - 1.5. Insert minimum five records in above tables.
 - 1.6. Create table named 'tblsub_allotment' with fields 'said, sid, tid, division, assignyear'
(Here sid and tid are foreign keys)
 - 1.7. Create table named 'tblstud_result' with fields 'rno, sub1, sub2, sub3, sub4, sub5'
 - 1.8. Insert minimum 10 records in above two tables.
2. Create module named 'db_query.py' and create user defined function named db_conn(db, query) where the connection is established with the database (db) and where the query is executed. (Write general code in this function for executing any query.)
3. Perform following tasks by importing 'db_query.py' file. Use db_conn() to get output of following queries:
 - 3.1. Display the details of all subjects.
 - 3.2. Display the details of male teachers.
 - 3.3. Display the subject allotment detail of 'Hiral' teacher.
 - 3.4. Display the student's result whose roll no is given by user.
 - 3.5. Display subject allotment detail of current year.
4. Create 'stud.csv' file which includes 'rno, name, sub1, sub2, sub3, sub4, sub5' fields. Retrieve data from above created tables.
5. Extract data from 'stud.csv' file and store it as dataframe. Perform following tasks:
 - 5.1. Display line chart which includes rollno on x axis and Sub1's marks on y axis.
 - 5.1.1. Give proper headings.
 - 5.1.2. Display grid.
 - 5.2. Display comparison line chart for Sub1, Sub2, Sub3, Sub4 and Sub5 with respect to Rollno.
 - 5.2.1. Give proper heading.
 - 5.2.2. Display legend.
 - 5.2.3. Display each line with different formatting.
 - 5.3. Display above two charts in one figure (Use subplot-columnwise)
6. Extract data of 'rno, name, sub1, sub2, sub3, sub4, sub5' directly from SQLite database and create dataframe using it (Use DataFrame()). And perform following tasks:

- 6.1. Draw bar chart for Sub5 with respect to Name
 - 6.1.1. Give proper headings
 - 6.2. Draw scatter chart for Sub1, Sub2, Sub3, Sub4, Sub5 with different formatting.
 - 6.2.1. Give proper headings.
 - 6.3. Draw histogram chart for Sub4's marks with proper heading.
 - 6.4. Draw histogram chart for Sub2 as per [0,5,10,15,20,25,30,35,40,45,50] intervals.
7. Perform following tasks:
- 7.1. Create excel file which includes following details:

Year	Daily	Education	Health	Entertainn	Total
2014	500	20000	15000	10000	45500
2015	1000	23000	5000	8000	37000
2016	1500	25000	12000	12000	50500
2017	800	10000	2000	15000	27800
2018	1700	15000	3000	10000	29700
2019	1500	23000	4000	5000	33500
2020	900	25000	12000	7000	44900
2021	700	26000	23000	12000	61700
 - 7.2. Extract Year, Daily, Educational, Health and Entertainment expenses related details as dataframe.
 - 7.3. Draw BAR chart which includes all expenses.
 - 7.3.1. Give proper heading and display legend to the top-right corner of the figure.
 - 7.3.2. Display grid.
8. The law of seatbelt related law was introduced on 31st Jan., 1983. Before and after accident-related data are available in "seatbelts.csv" file. Copy the file and perform following tasks:
- 8.1. Import seatbelts.csv file into python.
 - 8.2. Calculate standard deviation of drivers killed after the law came into effect.
 - 8.3. Calculate variance for "Front" and "Rear" before the law came into effect.
 - 8.4. Find out the maximum kilometre driven by the driver.
 - 8.5. Find out the mean of total numbers of drivers killed before the law came into effect.
 - 8.6. Find out the mean of total numbers of drivers killed after the law came into effect.
9. Import seatbelts.csv file's data of after law came into effect.
- 9.1. Display the Line chart for "Front" and "Rear" .
 - 9.2. Display appropriate heading and show the legend.
10. Import seatbelts.csv file
- 10.1. Display interval wise frequency graph which shows the VANKILLER's data in 20 intervals between 0 to 18 for before law came into effect.
 - 10.2. Display interval wise frequency graph which shows the VANKILLER's data in 20 intervals between 0 to 18 for after law came into effect.
 - 10.3. Display both graphs in one figure with vertical division. (Use Subplot)